

Joseph Bradshaw

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Personal profile

An enthusiastic final year Mechanical Engineering student currently studying at Aston University with a strong interest in fluid engineering, manufacturing processes and product development. Recognising the strength that diversity in teams brings, or when working on my own, I always establish positive working relationships by maintaining a high focus on my support and responsibility to my team peers. This includes planning, monitoring, prioritising and communicating my responsibilities and outputs as they progress. If setbacks arise, I approach them with determination and a solutions-focused mindset, using each challenge as an opportunity to develop further as an individual and as a team. It's in dynamic environments like these that I thrive, confidently switching between tasks and maintaining composure under pressure. I am able to communicate my ideas and those of my team members efficiently and with confidence, recognising the importance of learning from others while challenging groupthink.

Project experience

Aston University, Birmingham

September 2024 - Present

Throughout my 2 years at Aston University, I have undertaken multiple projects, involving the design, analysis and manufacture of engineering systems, including:

- Designing and prototyping a subsonic wind tunnel for use on small scale vehicle prototypes
- Designing and constructing a chassis for a small electric vehicle powered by a 12V DC battery.
- Designing and manufacturing a steel wind turbine designed for use by residents in rural areas in the Philippines
- Developing an easy-to-use pressure sensing insole for diabetic patients wishing to monitor their condition more discreetly, with a companion mobile app for viewing and tracking pressure readings.
- Remanufacturing a battery system to reduce weight and improve assembly/disassembly efficiency for industrial applications.

Alongside these projects, I have also carried out several research projects, including an academic report on vibration measurement technologies, and a lab report investigating the performance of a vapour compression refrigerator with varying condenser pressures and temperatures. I also carried out a materials and processes analysis on 6 water pumps, each with varying material choices and manufacturing processes. Each flange, impeller and pump body was examined under a microscope and tested to identify the materials and processes used, explaining the decisions of each.

Employment

Athlete, Nike Resorts World, Birmingham

June 2023 - Present

- I deliver a premium customer experience by proactively engaging customers, understanding their needs, and using strong product knowledge to recommend suitable solutions. This requires adaptation to multiple areas of the store on a rota basis, including sales floor, fitting rooms, tills, front-of-store meet and greet, and stockroom, maintaining high service standards in each. Colleagues are supported during busy periods by flexing across departments and reprioritising my tasks to ensure smooth store operations.
- I always ensure compliance with store processes and loss-prevention procedures, handling cash and stock responsibly while remaining vigilant to security risks, scams, and theft. I am also frequently trusted to take on managerial responsibilities when required, demonstrating reliability, strong judgement, and the ability to support team operations in the absence of management.

Event Access Coordinator, Desibeatz, Birmingham

January 2025 - Present

- Hosting events at multiple different venues with large crowds means quickly adjusting to the location and forming good relationships with the staff is vital to a safe event. Through efficient communication and teamwork we discuss the most effective position and procedure to take. I manage the crowd and scan tickets, maintaining a cordial relationship with customers and following security procedures where required, working alongside security staff to manage latecomers, employing conflict resolution and de-escalation when required.

Bank Receptionist, Midlands Medical Partnership, Jockey Road Medical Centre

September 2022 - April 2023

- I worked as a receptionist in a large GP partnership, developing strong administrative, planning and communication skills in a busy, customer-facing environment. I supported patients who were unwell or distressed, providing guidance and referring to appropriate services when required, all whilst managing challenging interactions with empathy, resolving issues efficiently and escalating where necessary.
- I maintained strict patient confidentiality throughout, handling high volumes of sensitive data accurately and in line with information governance policies. I demonstrated the ability to work to tight deadlines, ensure precise record-keeping, and manage competing priorities effectively.

Education

BEng, Mechanical Engineering

Aston University, Birmingham, 2024 to present

Achieved an overall score of 71% for the across 2 years of the course, on track to achieve a First.

A-levels

Bishop Vesey's Grammar School, Birmingham, 2022-2024

Maths, Biology and Chemistry - achieved grades of BBB

GCSEs

Bishop Vesey's Grammar School, Birmingham, 2017-2022

76 points across ten GCSE subjects including 8s in English Language and Maths

Additional experience, awards, interests and qualifications

- Introduction to Design, Jaguar Land Rover, February 2018 (two-day work experience programme to gain insight into engineering, design and production)
- 3 years experience using Solidworks
- Arduino, sensors and wiring
- ANSYS
- Workshop manufacture (machining, 3D printing), testing, data analysis and report writing
- Member of the Institution of Mechanical Engineers (iMechE) since November 2024.
- Illustrator for The Data Journalism Handbook, a textbook for journalism students
- Level 3 First Aid trained (May 2023)
- Full UK driving licence
- GCSE Design Technology Award for 2022
- Black belt (2nd Dan), Wado Ryu Karate
- I have played and continue to enjoy a range of sports including football, hockey, and gym
- Volunteer, Kombat Kids
- Volunteer at the Inspiring Minds Programme, Bishop Vesey's Grammar School, providing PSHE lessons to year 5 children in local primary schools across Birmingham
- Duke of Edinburgh Award Bronze, September 2020 (partial award due to COVID-19)
- References available on request